How-To Library

Informatica

Configuring Ports for Big Data Products 10.2

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Abstract

You can deploy a solution consisting of several Informatica® products to address your requirements to extract, process and report data and metadata from big data sources. To prevent conflicts between products, this article tells you which ports are established when you run the installer for each product.

Supported Versions

- Informatica Intelligent Data Lake 10.2
- Informatica Enterprise Information Catalog 10.2
- Informatica Data Integration Hub 10.2
- Informatica Big Data Management 10.2

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Overview

You can enable several Informatica products in a solution that works together to extract, process and report data and metadata.

When you install Informatica products, the installer utility opens specific ports by default. You must open other ports manually. This article tells you which ports are established when you run Informatica 10.2 product installers.

Informatica Domain Services

The Informatica domain includes several services that perform important roles in data extraction and processing.

For more information about application services, see the *Informatica 10.2 Application Services Guide*.

Application Services and Ports

Informatica domain services and application services in the Informatica domain have unique ports.

Informatica Domain

The following table describes the ports that you can set:

Port	Description
Service Manager port	Port number used by the Service Manager on the node. The Service Manager listens for incoming connection requests on this port. Client applications use this port to communicate with the services in the domain. The Informatica command line programs use this port to communicate to the domain. This is also the port for the SQL data service JDBC/ODBC driver. Default is 6006.
Service Manager Shutdown port	Port number that controls server shutdown for the domain Service Manager. The Service Manager listens for shutdown commands on this port. Default is 6007.
Informatica Administrator port	Port number used by Informatica Administrator. Default is 6008.
Informatica Administrator HTTPS port	No default port. Enter the required port number when you create the service. Setting this port to 0 disables an HTTPS connection to the Administrator tool.
Informatica Administrator shutdown port	Port number that controls server shutdown for Informatica Administrator. Informatica Administrator listens for shutdown commands on this port. Default is 6009.
Minimum port number	Lowest port number in the range of dynamic port numbers that can be assigned to the application service processes that run on this node. Default is 6014.
Maximum port number	Highest port number in the range of dynamic port numbers that can be assigned to the application service processes that run on this node. Default is 6114.

Analyst Service

The following table lists the default port associated with the Analyst Service:

Type	Default Port
Analyst Service (HTTP)	8085
Analyst Service (HTTPS)	No default port. Enter the required port number when you create the service.

Content Management Service

The following table lists the default port associated with the Content Management Service:

Type	Default Port
Content Management Service (HTTP)	8105
Content Management Service (HTTPS)	No default port. Enter the required port number when you create the service.

Data Integration Service

The following table lists the default port associated with the Data Integration Service:

Type	Default Port
Data Integration Service (HTTP proxy)	8080
Data Integration Service (HTTP)	8095
Data Integration Service (HTTPS)	No default port. Enter the required port number when you create the service.
Profiling Warehouse database	No default port. Enter the database port number.

Metadata Access Service

The following table lists the default port associated with the Metadata Access Service:

Type	Default Port
Metadata Access Service (HTTP)	7080
Metadata Access Service (HTTPS)	No default port. Enter the required port number when you create the service.

Metadata Manager Service

The following table lists the default port associated with the Metadata Manager Service:

Type	Default Port
Metadata Manager Service (HTTP)	10250
Metadata Manager Service (HTTPS)	No default port. Enter the required port number when you create the service.

PowerExchange® Listener Service

Use the same port number that you specify in the SVCNODE statement of the DBMOVER file.

If you define more than one Listener Service to run on a node, you must define a unique SVCNODE port number for each service.

PowerExchange Logger Service

Use the same port number that you specify in the SVCNODE statement of the DBMOVER file.

If you define more than one Listener Service to run on a node, you must define a unique SVCNODE port number for each service.

Web Services Hub Service

The following table lists the default port associated with the Web Services Hub Service:

Type	Default Port
Web Services Hub Service (HTTP)	7333
Web Services Hub Service (HTTPS)	7343

Data Integration Hub Port Numbers

The Data Integration Hub installer sets the default port numbers for the installation components. If another application uses the same port number as one of the installation components, a port conflict might prevent the component from running correctly or cause errors.

You can change the port numbers after installation. Before you start Data Integration Hub, verify that the port numbers do not conflict with other applications and change the port numbers in Data Integration Hub to prevent port conflicts.

The following table describes the default port numbers:

Description	Port Number
UDP multicast port that Data Integration Hub uses for internal communications.	18000
Operation Console shutdown port. Only required to be available on the machine where Data Integration Hub is installed.	18005
Port that the Operation Console uses for internal communications.	18050
Operation Console HTTP port. Required only if you use an HTTP port for the Operation Console.	18080
RMI port for Data Integration Hub server startup and shutdown.	18095
RMI ports that the Operation Console and PowerCenter workflows use to communicate with the Data Integration Hub server.	18095 and 10896
Port that the Data Integration Hub server uses for internal communications.	18100
Operation Console HTTPS port. Required only if you use an HTTPS port for the Operation Console.	18443
Port for the Data Integration Hub JNDI provider. This port is also the JMS listener port for the Data Integration Hub JMS Broker.	18616

Enterprise Information Catalog Ports

The following table lists the ports that Enterprise Information Catalog uses:

Description	Port
Ambari Agent TCP	8670
Ambari Metrics Collector TCP	6188, 60200
Ambari Server TCP	8080, 8440, 8441
Ambari Server UDP	59804
Catalog Service	9085
HDFS DataNode	50010
HDFS DataNode HTTP	50075
HDFS DataNode HTTPS	50475

Description	Port
HDFS DataNode IPC	8010
HDFS HTTPS	50470
HDFS JournalNode HTTP	8480
HDFS JournalNode HTTPS	8481
HDFS NameNode HTTP	50070
HDFS NameNode HTTPS	50470
HDFS NameNode RPC	8020
HDFS Secondary Name Node	50090
Informatica Cluster Service	9075
pgSQL	5432
YARN JobHistory Server	19888
YARN NodeManager	45454
YARN ResourceManager	8032
YARN ResourceManager Admin	8141
YARN ResourceManager ResourceTracker	8025
YARN ResourceManager Scheduler	8030
YARN ResourceManager Webapp	8088
YARN ResourceManager Webapp HTTPS	8090
YARN Timeline Server	10020
YARN Timeline Server Webapp	8188
YARN Timeline Server Webapp HTTPS	8190
Zookeeper	2181

Big Data Management Ports

The following sections list Hadoop components and default port numbers for various Hadoop distributions.

Note: Administrators can edit port number settings to use different ports. Check the ports that your Big Data Management implementation uses.

Amazon EMR Port Reference

The following table lists the ports associated with Amazon EMR:

Hadoop Component	Port
HBase master	60000
HBase master web	60010
HBase region server	60020
HDFS read/write	50010, 50020
HDFS namenode HTTP	50070
Hive metastore	9083
HiveServer	10000
JobTracker	8021
JobTracker web manager	50030
MapReduce Application Master	50100-50200
MapReduce JobHistory server	10020
MapReduce JobHistory server webapp	19888
NameNode	8020
ResourceManager	8032
ResourceManager webapp	8088
ResourceTracker	8031
Scheduler address	8030
Shuffle HTTP	13562
TaskTracker web management	50060
ZooKeeper	2181

IBM BigInsights Port Reference

The following table lists the default ports associated with IBM BigInsights:

Hadoop Component	Port
Ambari cluster management tool	8080
HBase master	60000

Hadoop Component	Port
HBase master web	60010
HBase region server	60020
HDFS read/write	50010, 50020
HDFS namenode HTTP	50070
Hive metastore	9083
HiveServer	10000
JobTracker	9001
JobTracker web manager	50030
MapReduce Application Master	50100-50200
MapReduce JobHistory server	10020
MapReduce JobHistory server webapp	19888
NameNode	9000
ResourceManager	8032
ResourceManager webapp	8088
ResourceTracker	8031
Scheduler address	8030
Shuffle HTTP	13562
TaskTracker web management	50060
ZooKeeper	2181

Cloudera Port Reference

The following table lists the ports associated with Cloudera:

Hadoop Component	Port
Cloudera Manager	7180
HBase master	60000
HBase master web	60010
HBase region server	60020

Hadoop Component	Port
HDFS read/write	50010, 50020
HDFS namenode HTTP	50070
Hive metastore	9083
HiveServer	10000
JobTracker	8021
JobTracker web manager	50030
MapReduce Application Master	50100-50200
MapReduce JobHistory server	10020
MapReduce JobHistory server webapp	19888
NameNode	8020
ResourceManager	8032
ResourceManager webapp	8088
ResourceTracker	8031
Scheduler address	8030
Shuffle HTTP	13562
TaskTracker web management	50060
ZooKeeper	2181

HDInsight Port Reference

The following table lists the ports associated with HDInsight:

Hadoop Component	Port
Ambari cluster management tool	8080
HBase master	60000
HBase master web	60010
HBase region server	60020
HDFS read/write	50010, 50020
HDFS namenode HTTP	50070

Hadoop Component	Port
Hive metastore	9083
HiveServer	10000
JobTracker	9001
JobTracker web manager	50030
MapReduce Application Master	50100-50200
MapReduce JobHistory server	10020
MapReduce JobHistory server webapp	19888
NameNode	9000
ResourceManager	8032
ResourceManager webapp	8088
ResourceTracker	8031
Scheduler address	8030
Shuffle HTTP	13562
TaskTracker web management	50060
ZooKeeper	2181

Hortonworks Port Reference

The following table lists the ports associated with Hortonworks:

Hadoop Component	Port
Ambari cluster management tool	8080
HBase master	60000
HBase master web	60010
HBase region server	60020
HDFS read/write	50010, 50020
HDFS namenode HTTP	50070
Hive metastore	9083
HiveServer	10000

Hadoop Component	Port
JobTracker	8021
JobTracker web manager	50030
MapReduce Application Master	50100-50200
MapReduce JobHistory server	10020
MapReduce JobHistory server webapp	19888
NameNode	8020
ResourceManager	8032
ResourceManager webapp	8088
ResourceTracker	8031
Scheduler address	8030
Shuffle HTTP	13562
TaskTracker web management	50060
ZooKeeper	2181

MapR Port Reference

The following table lists the ports associated with MapR:

Hadoop Component	Port	Notes
CLDB	7222	
CLDB JMX monitor	7220	
CLDB web port	7221	
HBase master	16000	
HBase master web	16010	
HBase region server	16020	
Hive metastore	9083	
HDFS namenode HTTP	50070	
HiveServer	10000	
JobTracker	9001	

Hadoop Component	Port	Notes
JobTracker web manager	50030	
MapR Control System (MCS) cluster management tool	8443	
MapReduce Application Master	50100-50200	
MapReduce JobHistory server	10020	YARN only
MapReduce JobHistory server web app	19888	YARN only
ResourceManager	8032	YARN only
ResourceManager web app	8088	YARN only
Secure Resource Manager web app (HTTPS)	8090	
ResourceTracker	8031	YARN only
Scheduler address	8030	YARN only
Shuffle HTTP	13562	
TaskTracker web management	50060	
ZooKeeper	5181	

Hadoop Cluster Port Reference

A Hadoop cluster requires some unique port numbers.

Hadoop Cluster Management Tool Ports

Use the following table to find the default URL and ports for the cluster management web page for each Hadoop distribution:

Hadoop Distribution	Console	Default Cluster Management Tool Port
Amazon EMR	AWS Console	Not applicable. Amazon EMR is managed within its own website at https://aws.amazon.com/console/.
IBM BigInsights	Ambari	8080
Cloudera	Cloudera Manager	7180
HDInsight	Ambari	8080
Hortonworks	Ambari	8080
MapR	MapR Control System	8443

Blaze Engine Port Reference

The Blaze engine uses ports for the Blaze Job Monitor and the Blaze services.

Blaze Job Monitor

The following table lists the ports for the Blaze Job Monitor:

Port Name	Default Port Number
HTTP	9080
JSF	9090

Blaze Services

The following table lists the ports for Blaze services:

Port Name	Default Port Number
Minimum and Maximum Port Range for Hadoop Connection	12300 - 12600

Developer Tool Port Reference

The following table lists the ports for the Developer tool:

Port Name	Default Port Number
CLDB	7222 - MapR only
HBase master	60000
HBase region server	60200
HiveServer2 thrift	10000

Port Name	Default Port Number
NameNode RPC	8020 - except MapR
ZooKeeper client	2181, 5181

Intelligent Data Lake Ports

When you install Intelligent Data Lake, the installer opens the following ports.

Data Preparation Service

The following table lists the ports you can configure for the Data Preparation Service:

Description	Default Port Number
HTTP	8099
Rules Server Port	8098
Solr Port	8983

Intelligent Data Lake Service

The following table lists the port you can configure for the Intelligent Data Lake Service:

Description	Default Port Number
HTTP	9045

Author

Mark Pritchard